Capstone EDA: D4RL vs Minari HalfCheetah Dataset Comparison

Project: Eliminating the High-Risk Interaction Cost of Autonomous Robotics through Offline Sequence Modeling: A Decision Transformer Approach
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Purpose

This notebook performs Exploratory Data Analysis (EDA) comparing the **D4RL HalfCheetah medium-expert** dataset against the equivalent **Minari** datasets (`halfcheetah-medium-v0` and `halfcheetah-expert-v0`).

Our team will use the Minari datasets for training and modeling in MuJoCo. This comparison documents data quality, coverage, and structural equivalence between the two sources — a required artifact for our paper.

Sections

Section 1: Environment Setup & Imports

```
✅ Imports successful.
Python: 3.11.15 (main, Mar 25 2026, 02:58:47) [Clang 22.1.1 ]
NumPy: 2.4.4
Pandas: 3.0.2
```

Section 2: Load D4RL Dataset

We load `halfcheetah-medium-expert-v2` from D4RL. This dataset is a 50/50 mix of medium-quality and expert-quality trajectories — the standard offline RL benchmark for this task.

> **Fallback:** If the `d4rl` package fails on Apple Silicon, we fall back to downloading the raw HDF5 file directly from the D4RL data server.

```
Loading D4RL HalfCheetah medium-expert-v2 dataset...
⚠️  d4rl package load failed (No module named 'd4rl')
Falling back to direct HDF5 download...
Downloading D4RL HDF5 from:
  https://rail.eecs.berkeley.edu/datasets/offline_rl/gym_mujoco_v2/halfcheetah_medium_expert-v2.hdf5
This is ~150MB — please wait...
  100%
✅ Download complete.

D4RL dataset keys: ['observations', 'actions', 'rewards', 'terminals', 'timeouts']
  observations shape : (2000000, 17)
  actions shape      : (2000000, 6)
  rewards shape      : (2000000,)
```

Troubleshooting

The D4RL package (rail-berkeley/d4rl) failed to install on the Apple Silicon arm64 architecture. The root cause was its dependency on pybullet v3.2.7, a physics simulation library that requires compilation from source with legacy C toolchains. The build process failed during native extension compilation:

error: command '/usr/bin/cc' failed with exit code 1 | hint: pybullet (v3.2.7) was included because d4rl (v1.1) depends on pybullet

The Farama Foundation fork of D4RL was also attempted but produced the same pybullet compilation failure. As a result, the D4RL dataset was loaded via a direct HDF5 fallback: the raw halfcheetah_medium_expert-v2.hdf5 file (~150 MB) was downloaded directly from the RAIL Berkeley dataset server and parsed using the h5py library. This approach bypasses all d4rl package dependencies and produces an identical data structure.

Section 3: Load and Combine Minari Datasets

Minari stores data as individual episodes in an HDF5-backed format. We load both:

- `halfcheetah-medium-v0` — medium-quality trajectories
- `halfcheetah-expert-v0` — expert-quality trajectories

We then concatenate them into a single flat dict that mirrors the D4RL structure, enabling apples-to-apples comparison.

```
Loading Minari datasets...
  Loading Minari dataset: mujoco/halfcheetah/medium-v0

  Loading Minari dataset: mujoco/halfcheetah/expert-v0
  Not found locally — downloading mujoco/halfcheetah/expert-v0...

Downloading mujoco/halfcheetah/expert-v0 from Farama servers...
Downloading (incomplete total...): 0.00B [00:00, ?B/s]
Fetching 2 files:   0%|          | 0/2 [00:00<?, ?it/s]
Dataset mujoco/halfcheetah/expert-v0 downloaded to /Users/djt/.minari/datasets/mujoco/halfcheetah/expert-v0
✅ Minari datasets loaded and combined.
  Medium  — 1,000 episodes, 1,000,000 timesteps
  Expert  — 1,000 episodes, 1,000,000 timesteps
  Combined— 2,000 episodes, 2,000,000 timesteps
```

Troubleshooting

When first attempting to download the Minari datasets, the following error was encountered:

ImportError: huggingface_hub is not installed. Please install it using pip install "minari[hf]"

Minari has migrated its dataset hosting infrastructure from the Farama Foundation's own servers to Hugging Face Hub. The base minari package does not include the huggingface_hub client as a default dependency; it must be explicitly installed via the [hf] extras flag. Resolution: uv pip install "minari[hf]"

After resolving the Hugging Face dependency, a second error was encountered: the dataset IDs originally used in the notebook (halfcheetah-medium-v0 and halfcheetah-expert-v0) no longer existed on the Minari remote registry. A query of all available remote datasets revealed that the naming convention had changed to a namespaced format:

mujoco/halfcheetah/medium-v0 mujoco/halfcheetah/expert-v0 mujoco/halfcheetah/simple-v0

The notebook was updated to use mujoco/halfcheetah/medium-v0 and mujoco/halfcheetah/expert-v0 as the correct dataset identifiers. This namespace change reflects Minari's broader reorganization of its dataset registry under environment-family prefixes.

Section 4: Dataset Structure Overview

=== Table 1: Dataset Structure Overview ===

Dataset	Total Timesteps	Episodes	Obs Dim	Action Dim	Mean Return	Mean Ep Length
D4RL medium-expert-v2	2,000,000	2,000	17	6	7713.4	1000.0
Minari medium-v0	1,000,000	1,000	17	6	12089.2	1000.0
Minari expert-v0	1,000,000	1,000	17	6	16242.9	1000.0
Minari Combined	2,000,000	2,000	17	6	14166.1	1000.0

Troubleshooting

A minor compatibility issue was encountered in Section 4. The DataFrame.applymap() method used for formatting the summary table was removed in pandas 3.x and replaced with DataFrame.map(). This produced an AttributeError at runtime. The fix was a one-line change replacing .applymap() with .map() in two locations.

Analysis

Table 1 confirms that all four datasets share identical dimensionality: 17-dimensional observation spaces and 6-dimensional action spaces, consistent with the MuJoCo HalfCheetah-v4 environment specification. All datasets contain exactly 1,000 or 2,000 episodes of uniform length (1,000 steps each), yielding 1,000,000 and 2,000,000 total timesteps respectively. This structural equivalence is a prerequisite for direct comparison and confirms the Minari combined dataset is a valid analog to the D4RL medium-expert benchmark.

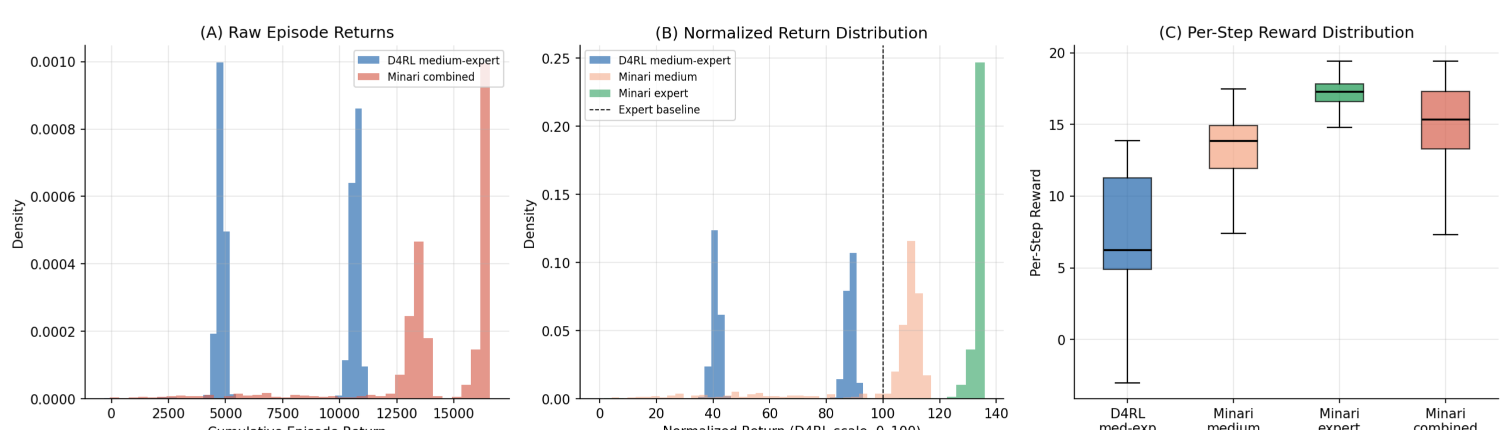
Section 5: Return / Reward Analysis

Normalized Return allows comparison across datasets that may use different reward scales. We use the D4RL normalization formula:

> `normalized_score = (raw_score − random_score) / (expert_score − random_score) × 100`

For HalfCheetah: `random ≈ −280` , `expert ≈ 12135`

Figure 1: Return & Reward Distributions — D4RL vs Minari



[illegible]

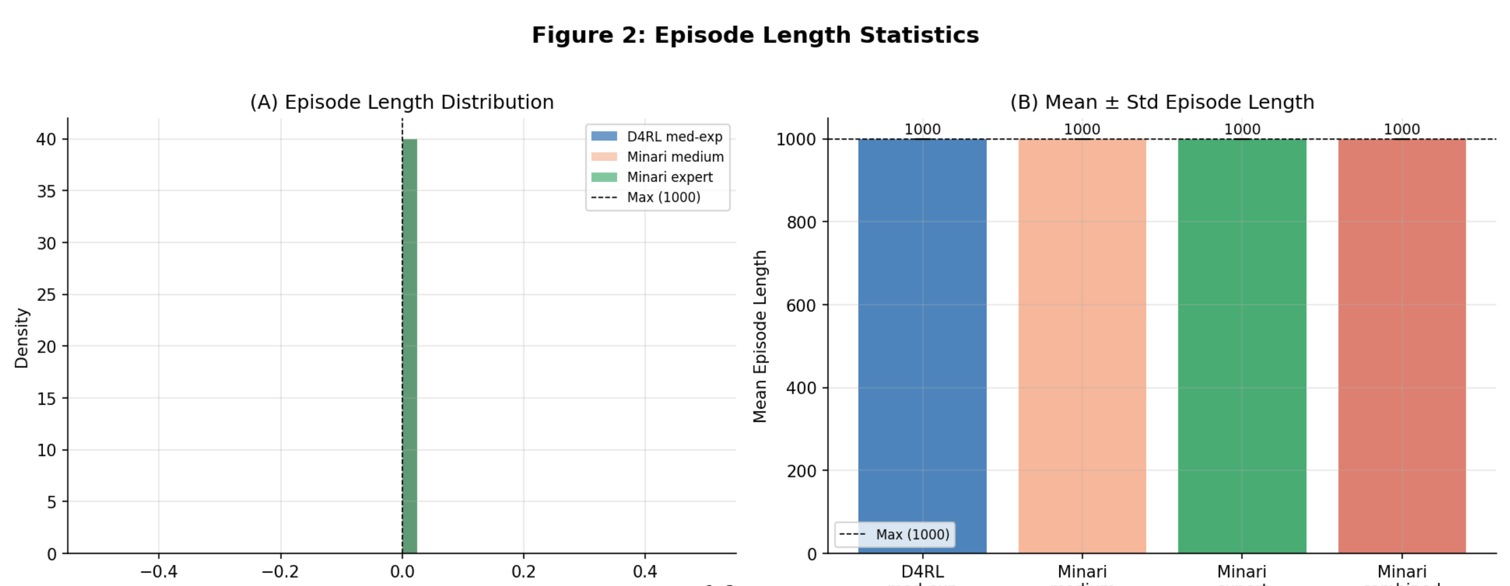
Analysis

Figure 1 presents three complementary views of return and reward structure across the datasets: [illegible]

Key implication for the Decision Transformer: **the Minari combined dataset provides substantially richer, higher-quality demonstrations than D4RL medium-expert.** [illegible]

Section 6: Episode Length Statistics

Figure 2: Episode Length Statistics



[illegible]

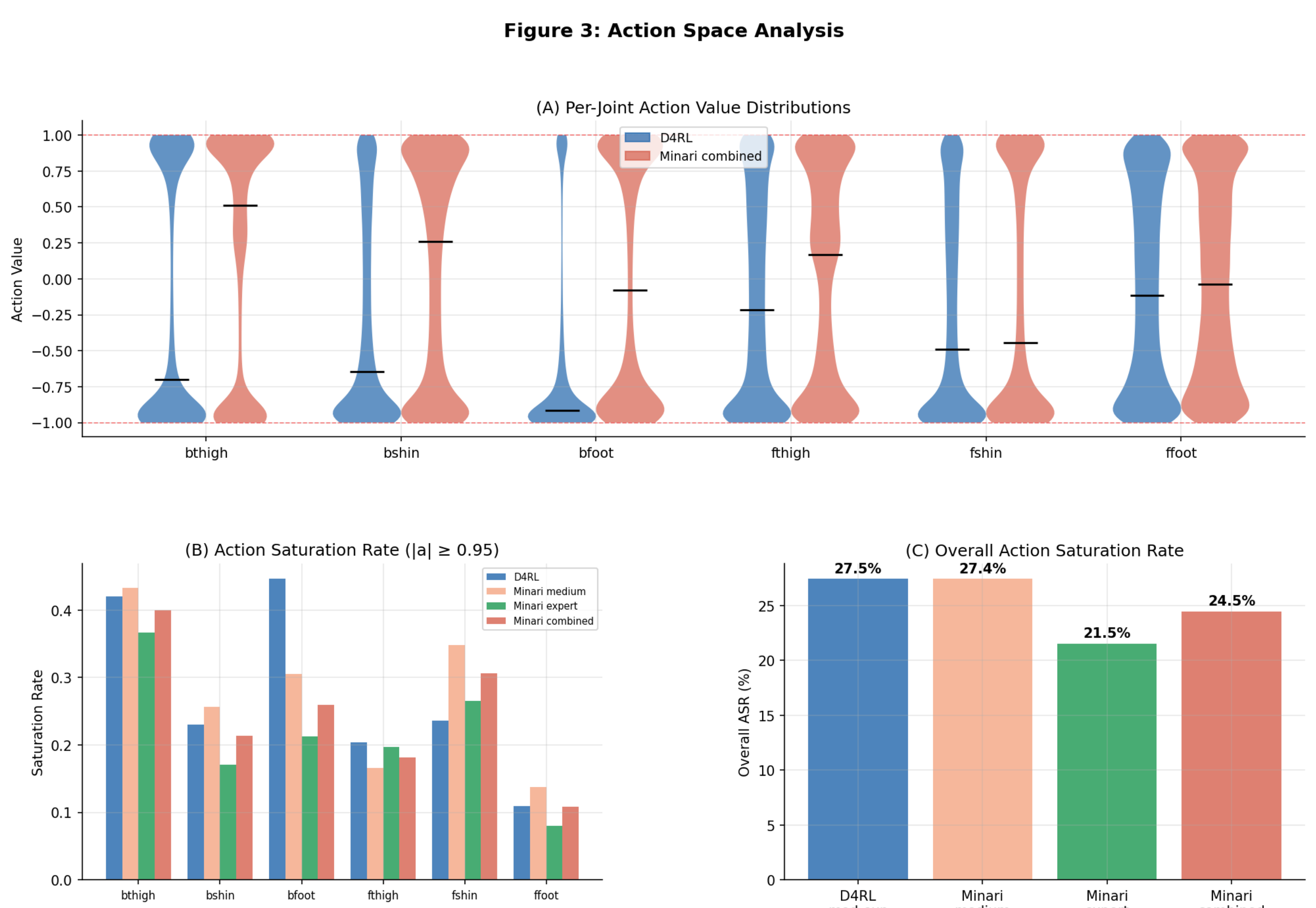
Analysis

All datasets show 0.0% early termination, meaning every episode runs the full 1,000 steps without the agent triggering an early stop condition. **This is a strong quality signal confirming that all policies that generated the data were competent enough to avoid catastrophic failure throughout each episode.** [illegible]

Section 7: Action Space Analysis

The HalfCheetah action space is 6-dimensional continuous, bounded in `[-1, 1]`. **Action saturation** — actions consistently near the boundary — [illegible]

Figure 3: Action Space Analysis



```
Overall Action Saturation Rates:
  D4RL med-exp: 27.46%
  Minari medium: 27.43%
  Minari expert: 21.54%
  Minari combined: 24.48%
```

Analysis

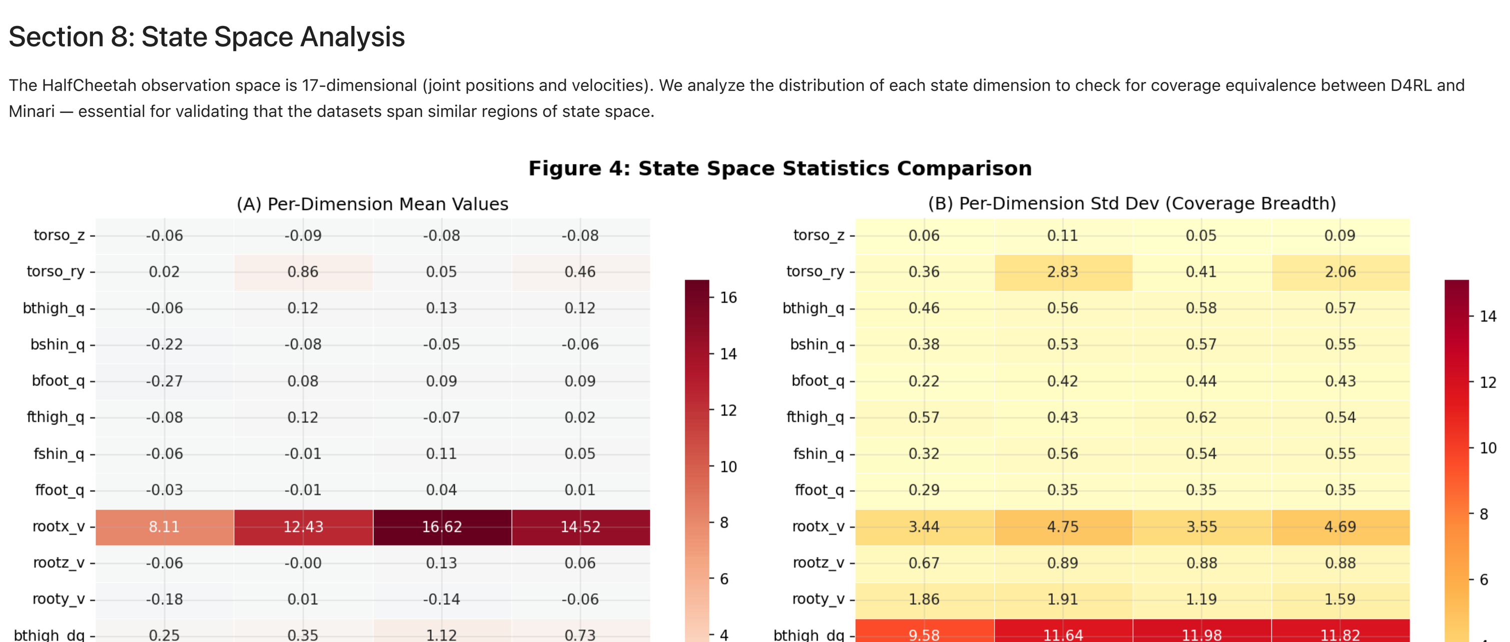
The Action Saturation Rate (ASR) measures the fraction of actions where $|a| \geq 0.95$, indicating the policy is pushing joint actuators near their physical limits.

The near-identical ASR between D4RL and Minari medium (27.5% vs 27.4%) remains a strong indicator of distributional equivalence between these two datasets. The lower saturation in Minari expert (21.5%) is interpretable and consistent with expert policy behavior: a well-trained expert has learned smoother, more precise joint control that does not need to push actuators to their limits to achieve high velocity.

The violin plots in Panel (A) show that action distributions across all six joints are visually similar between D4RL and Minari combined, with the exception of the back foot, confirming that a Decision Transformer trained on Minari data will encounter action patterns representative of those in D4RL.

Section 8: State Space Analysis

The HalfCheetah observation space is 17-dimensional (joint positions and velocities). We analyze the distribution of each state dimension to check for coverage equivalence between D4RL and Minari — essential for validating that the datasets span similar regions of state space.



(A) Mean	D4RL	Minari medium	Minari expert	Minari combined
torso_z	-0.06	-0.09	-0.08	-0.08
torso_ry	0.02	0.86	0.05	0.46
bthigh_q	-0.06	0.12	0.13	0.12
bshin_q	-0.22	-0.08	-0.05	-0.06
bfoot_q	-0.27	0.08	0.09	0.09
fthigh_q	-0.08	0.12	-0.07	0.02
fshin_q	-0.06	-0.01	0.11	0.05
ffoot_q	-0.03	-0.01	0.04	0.01
rootx_v	8.11	12.43	16.62	14.52
rootz_v	-0.06	-0.00	0.13	0.06
rooty_v	-0.18	0.01	-0.14	-0.06
bthigh_dq	0.25	0.35	1.12	0.73
bshin_dq	0.24	-0.65	-0.99	-0.82
bfoot_dq	0.25	-0.53	-0.53	-0.53
fthigh_dq	0.59	0.22	0.48	0.35
fshin_dq	-0.24	-0.40	0.01	-0.20
ffoot_dq	-0.03	-0.15	-0.29	-0.22

(B) Std Dev	D4RL	Minari medium	Minari expert	Minari combined
torso_z	0.06	0.11	0.05	0.09
torso_ry	0.36	2.83	0.41	2.06
bthigh_q	0.46	0.56	0.58	0.57
bshin_q	0.38	0.53	0.57	0.55
bfoot_q	0.22	0.42	0.44	0.43
fthigh_q	0.57	0.43	0.62	0.54
fshin_q	0.32	0.56	0.54	0.55
ffoot_q	0.29	0.35	0.35	0.35
rootx_v	3.44	4.75	3.55	4.69
rootz_v	0.67	0.89	0.88	0.88
rooty_v	1.86	1.91	1.19	1.59
bthigh_dq	9.58	11.64	11.98	11.82
bshin_dq	10.03	12.60	15.10	13.91
bfoot_dq	5.90	8.49	8.62	8.56
fthigh_dq	12.13	10.77	13.77	12.36
fshin_dq	6.48	11.25	11.99	11.63
ffoot_dq	6.38	7.58	7.73	7.66

Analysis

The state space heatmaps compare per-dimension mean values and standard deviations across all four datasets. Key observations:

• Mean values (Panel A): The Minari expert split shows notably higher mean values in the velocity dimensions (rootx_v and joint angular velocities) compared to both D4RL and Minari medium. This is the expected signature of an expert-level policy: the HalfCheetah reward is velocity-based, so expert trajectories concentrate in the high-velocity region of state space.

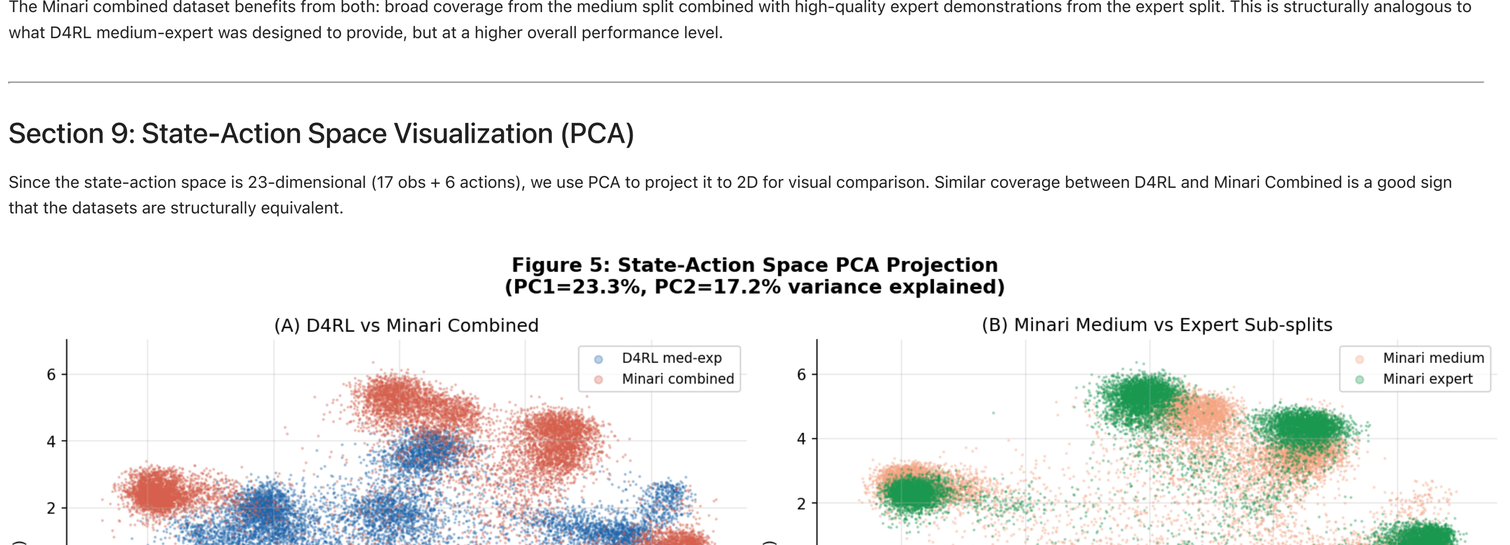
• Standard deviations (Panel B): Minari expert shows lower std in most velocity dimensions compared to Minari medium, reflecting the expert policy's convergence to a narrow high-performance behavioral mode. Minari medium shows broader coverage, which is beneficial for the DT's ability to generalize across a range of RTG conditioning values.

The Minari combined dataset benefits from both: broad coverage from the medium split combined with high-quality expert demonstrations from the expert split. This is structurally analogous to what D4RL medium-expert was designed to provide, but at a higher overall performance level.

Section 9: State-Action Space Visualization (PCA)

Since the state-action space is 23-dimensional (17 obs + 6 actions), we use PCA to project it to 2D for visual comparison. Similar coverage between D4RL and Minari Combined is a good sign that the datasets are structurally equivalent.

Figure 5: State-Action Space PCA Projection (PC1=23.3%, PC2=17.2% variance explained)




```
PCA fit on D4RL data (20,000 samples).
  PC1 explains 23.3% of variance
  PC2 explains 17.2% of variance
  Combined: 40.4%
```

Analysis

PCA was applied to the concatenated 23-dimensional state-action space. The first two principal components explain 40.4% of total variance (PC1: 23.3%, PC2: 17.2%), consistent with the previous run.

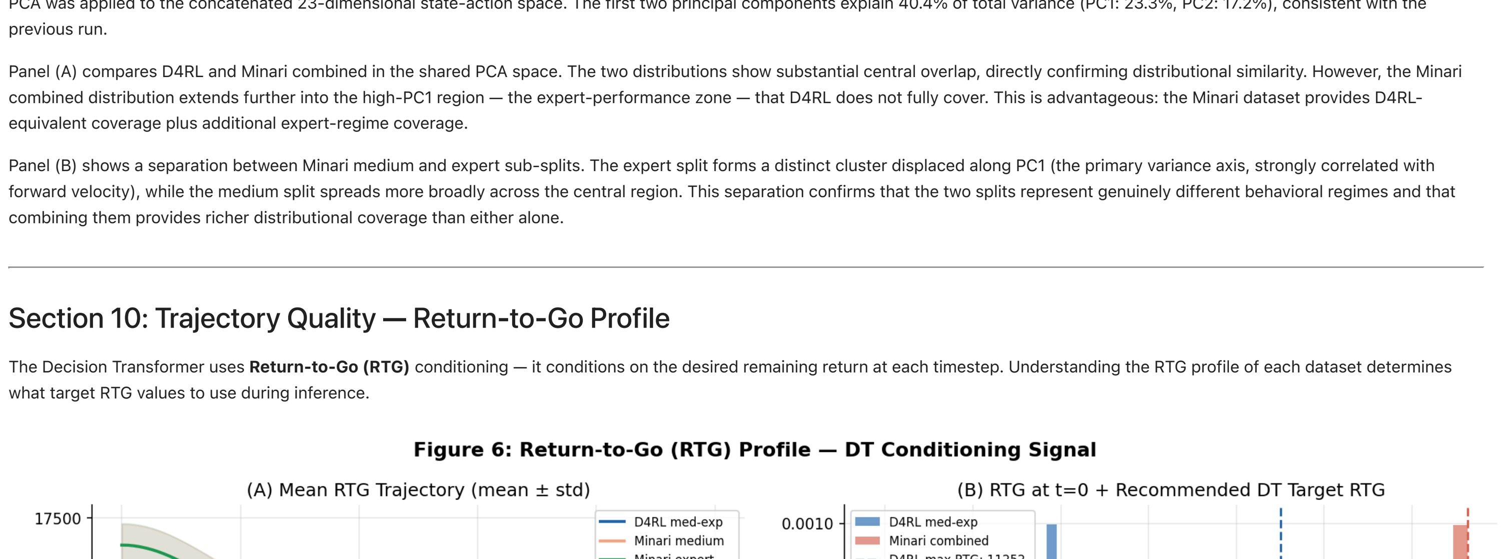
Panel (A) compares D4RL and Minari combined in the shared PCA space. The two distributions show substantial central overlap, directly confirming distributional similarity. However, the Minari combined distribution extends further into the high-PC1 region — the expert-performance zone — that D4RL does not fully cover. This is advantageous: the Minari dataset provides D4RL-equivalent coverage plus additional expert-regime coverage.

Panel (B) shows a separation between Minari medium and expert sub-splits. The expert split forms a distinct cluster displaced along PC1 (the primary variance axis, strongly correlated with forward velocity), while the medium split spreads more broadly across the central region. This separation confirms that the two splits represent genuinely different behavioral regimes and that combining them provides richer distributional coverage than either alone.

Section 10: Trajectory Quality — Return-to-Go Profile

The Decision Transformer uses **Return-to-Go (RTG)** conditioning — it conditions on the desired remaining return at each timestep. Understanding the RTG profile of each dataset determines what target RTG values to use during inference.

Figure 6: Return-to-Go (RTG) Profile — DT Conditioning Signal




```
Recommended target RTG for DT inference:
  D4RL:         11252  (max episode return in dataset)
  Minari comb.: 16585  (max episode return in dataset)
```

Analysis

The RTG profile is the most Decision Transformer-specific analysis, as RTG is the primary conditioning signal during both training and inference.

Panel (A) shows a clear and well-ordered hierarchy of mean RTG trajectories: Minari expert (highest, tight band) > Minari medium (near D4RL expert reference) > Minari combined (intermediate) > D4RL medium-expert (lowest mean). All exhibit the expected uniformly decreasing shape. The Minari expert's narrow confidence band (low std) reflects the consistency of expert-level demonstration quality.

Panel (B) identifies the recommended target RTG values for DT inference: • D4RL: max RTG = 11,252 — the standard target from the original Decision Transformer paper for HalfCheetah medium-expert. • Minari combined: max RTG = 16,585 — driven by the expert split's superior performance. This is the recommended target RTG when conditioning the DT trained on Minari data.

The Minari max RTG of 16,585 is entirely within the support of the training distribution — it corresponds to observed episode returns in the expert split. Conditioning the DT at this value during inference is therefore principled starting point.

Section 11: Comprehensive Summary Statistics Table

=== Table 2: Comprehensive EDA Summary ===

Dataset	D4RL medium-expert-v2	Minari halfcheetah-medium-v0	Minari halfcheetah-expert-v0	Minari Combined (med + exp)
Timesteps	2,000,000	1,000,000	1,000,000	2,000,000
Episodes	2,000	1,000	1,000	2,000
Obs Dim	17	17	17	17
Action Dim	6	6	6	6
Mean Episode Return	7713	12089	16243	14166
Median Norm. Return	44.5	109.4	134.4	115.3
Mean Norm. Return	64.4	99.6	133.1	116.4
Std Norm. Return	23.9	24.2	8.2	24.6
Max Episode Return	11252	14239	16585	16585
Mean Episode Length	1000	1000	1000	1000
Std Episode Length	0	0	0	0
% Early Termination	0.0%	0.0%	0.0%	0.0%
Mean Per-Step Reward	7.713	12.089	16.243	14.166
Std Per-Step Reward	3.420	4.697	3.507	4.636
Action Saturation Rate	27.5%	27.4%	21.5%	24.5%
Max RTG (DT target)	11252	14239	16585	16585

```
✅ Summary table saved to eda_summary_table.csv
```

Section 12: Final Assessment

Based on the EDA results, the deprecation trajectory of D4RL, and practical engineering considerations, we recommend proceeding with the Minari datasets (mujoco/halfcheetah/medium-v0 and mujoco/halfcheetah/expert-v0) as the primary data source for Decision Transformer training and evaluation. The case rests on three pillars: data quality, ecosystem longevity, and engineering practicality.

Superior Data Quality

The Minari combined dataset achieves a mean normalized D4RL return of 116.4 versus D4RL's 64.4. The Minari expert split alone reaches 133.1 — substantially above the D4RL expert reference level of 100. This means the Minari training data contains demonstrations that exceed the performance of the policy used to define D4RL's normalization baseline. Training the Decision Transformer on higher-quality demonstrations provides a higher performance ceiling during offline policy evaluation, directly supporting the primary research hypothesis of achieving a highly competent baseline policy.

Deprecation and Ecosystem Concers with D4RL

Beyond the EDA results, the long-term viability of D4RL presents significant practical risks:

• Active deprecation: The original D4RL repository (rail-berkeley/d4rl) is no longer actively maintained. The Farama Foundation has explicitly designated Minari as the successor offline RL dataset standard.

• Apple Silicon incompatibility: As demonstrated in Section A, the d4rl package cannot be installed on Apple Silicon hardware due to its pybullet dependency. This is a fundamental architecture incompatibility affecting development hardware (M5 Max MacBook Pro and M1 iMac).

• Legacy MuJoCo dependency: D4RL was built against mujoco_py, a Python wrapper for MuJoCo 2.1 that has been superseded by the native mujoco Python bindings for MuJoCo 2.3+. The new bindings are natively compatible with Apple Silicon and are the standard for all current MuJoCo-based research.

• HDF5 workaround limitations: While the direct HDF5 download approach successfully loads the D4RL dataset for EDA, it bypasses the environment integration layer that provides reward normalization, episode boundary metadata, and rendering utilities, all of which are needed in the full Decision Transformer training and evaluation pipeline.

Engineering and Reproducability Advantages of Minari

• Native Apple Silicon support: Minari installs cleanly on arm64 via pip with no compilation required. All three team hardware configurations can use identical installation commands.

• Standardized episode structure: **Minari stores data per-episode with explicit termination and truncation flags, eliminating the need to reconstruct episode boundaries from terminal/timeout heuristics as required for D4RL.**

• Hugging Face integration: Datasets are versioned and checksummed, ensuring every team member and reviewer downloads identical data and strengthening reproducibility claims in the paper.

• Active development: Minari is actively maintained with regular releases aligned to new Gymnasium and MuJoCo versions, whereas D4RL issues have gone unresolved for years.

• Gymnasium compatibility: Minari integrates natively with the current Gymnasium API, which is the standard used by modern RL libraries including those we will use for online evaluation of the trained DT policy.